

Chapter 4

Nutrition During Lactation



Objectives

At the end of the this session you will be able to:

- Explain the physiology of lactation.
- Discuss the human milk composition.
- Identify composition of breast milk
- Describe importance of breast feeding
- Identify the energy and nutrient requirements during Lactation.

Introduction

- ❖ The health care system, the workplace, and the community can either hinder or facilitate the initiation and continuation of breastfeeding.
- ❖ Health care professionals who wish to manage and promote breastfeeding should understand the physiology of lactation, the composition of human milk, and the benefits to mothers and infants.

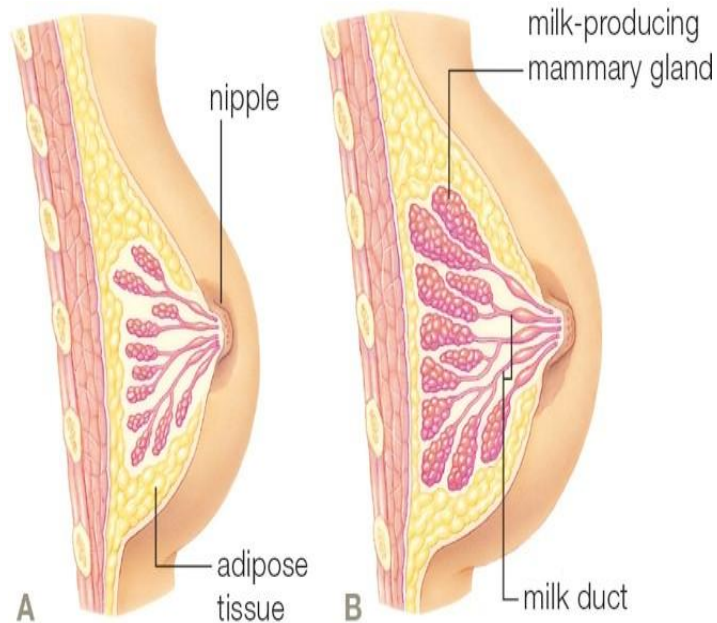
Key Nutrition Concept

1. Human milk is the best food for newborn infants for the first year of life or longer.
2. Maternal diet does not significantly alter the protein, carbohydrate, fat and major mineral composition of breast milk, but it does affect the fatty acid profile and the amounts of some vitamins and trace minerals.
3. When maternal diet is **inadequate**, the quality of the milk is preserved over the quantity for the majority of nutrients.

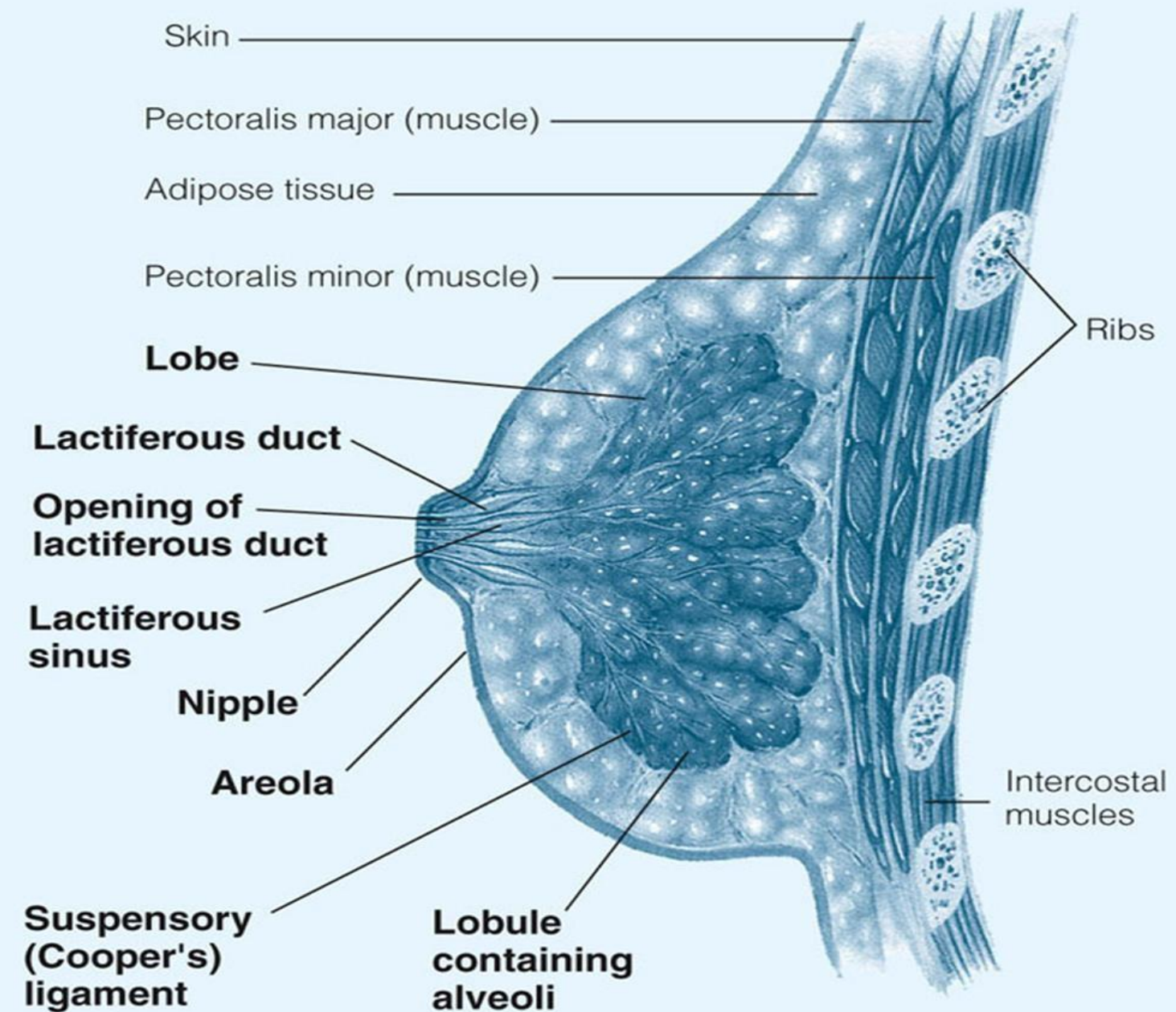
Lactation Physiology

■ Functional units of the mammary gland

- ✓ **alveoli**: rounded or oblong shaped cavity present in breast
- ✓ **Secretory cells**: responsible for secreting milk
- ✓ **Myoepithelial cells**: surround secretory cells and contract to cause milk ejection into ducts



Breast of a Lactating Female



Breast Development

- ❖ Begins at puberty
- ❖ Mammary glands remain dormant until pregnancy; grow during late pregnancy
- ❖ start to produce colostrom in late pregnancy
 - ❖ yelowsh liquid
 - ❖ High in protein, micronutrients, immunoglobulins
 - ❖ Replaced by milk after few days

Hormonal Control of Lactation

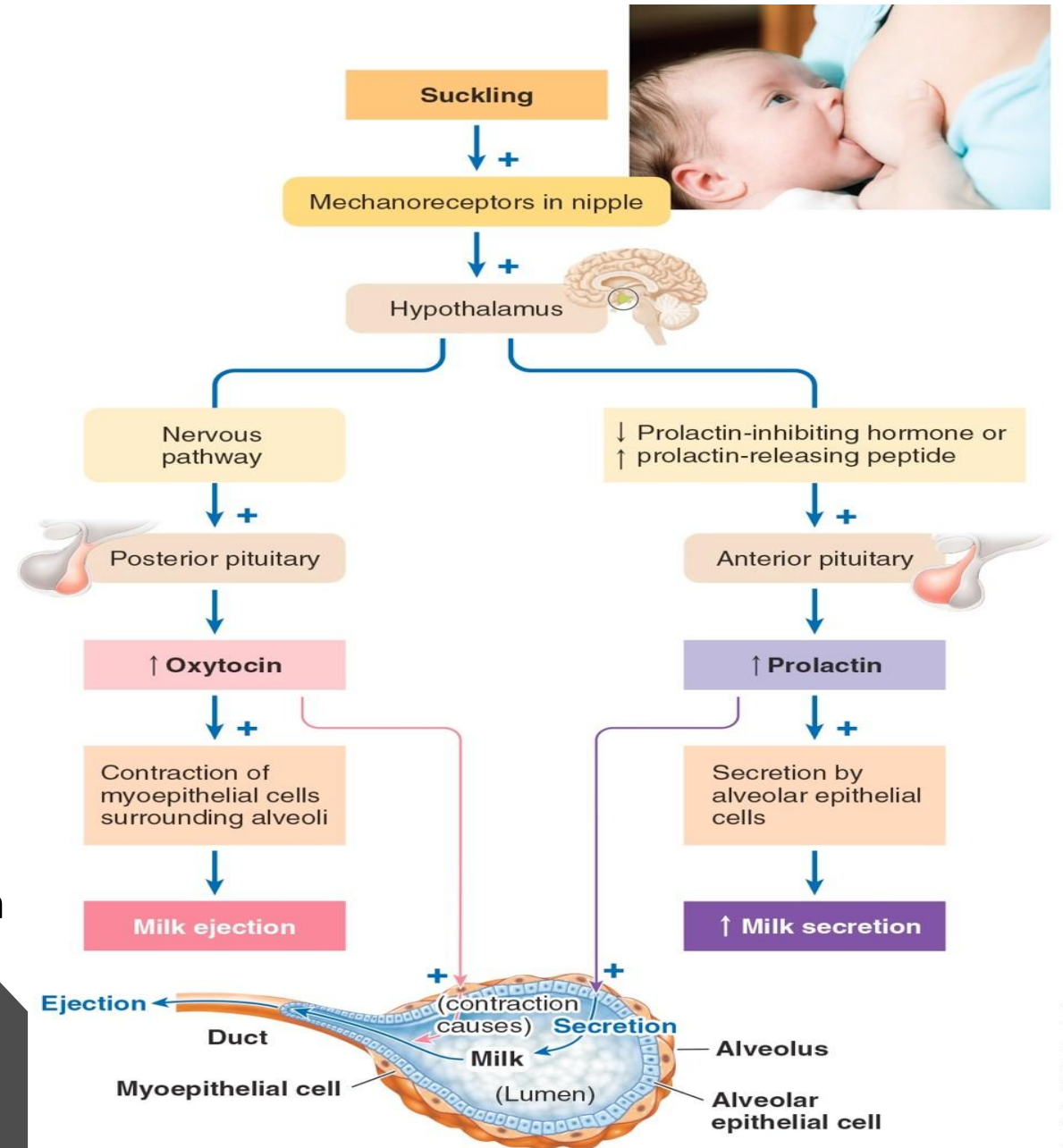
- **Prolactin**

- Stimulates milk production
- inhibit ovulation
- Released in response to suckling,
 - levels double with suckling

- **Oxytocin**

- Stimulates letdown/ ejection of milk from milk gland (acinus) into the milk ducts.
- Tingling of the breast may occur corresponding to contractions in milk duct
- Causes uterus to contract, seal blood vessels, & shrink in size

The letdown reflex



Physiology of Lactation

Lactogenesis

- ❖ it is regulation of breast milk secretion (production); it occurs in three stages.
- ✓ **Lactogenesis I** (begin in the third trimester) milk formation begins
- ✓ **Lactogenesis II**—(begins 2-5 days after birth) increased blood flow to breast; milk “comes in”
- ✓ **Lactogenesis III**—(begins at ~10 day after birth) milk composition is stable

Human Milk Composition

- Human milk is the only food needed by the majority of infants for approximately 6 months.
 - Human milk have hundreds of identified components.
 - The composition of milk is designed not only to **nurture**, but also to **protect infants from infectious and certain chronic diseases**.
- Depending upon the stage of lactation, breast milk is designated as:
1. *Colostrum* (3–5 days post partum)
 2. *Transitional milk* (5–15 days)
 3. *Mature milk* (secreted after 15 days or 2 weeks of infant birth).

Human Milk Composition

- ❖ **87% H₂O**—no need for extra water
- ❖ **Nutrient composition changes**
 - ✓ **With mother's diet**
 - Malnourished mother = low nutrient content
 - Taste changes depending on mother's diet, holds infant interest
 - ✓ **With time**
 - **Within feeding:**
 - Fore milk/first milk**: less fat, satisfies thirst
 - Hind milk**: higher fat, more satiating
 - ✓ **As lactation progresses**
 - **Protein: Whey: casein ratio** makes breastmilk more digestible
 - **90:10** in **early lactation**; **80:20** average; **60:40** late lactation



Human Milk Composition...

❖ Colostrum

- ✓ The first milk
- ✓ Is a thick, often yellow fluid produced (days 1–3 after infant birth).
- ✓ It provides about 580–700 kcal/l and is higher in protein and lower in carbohydrate & fat than mature milk.
- ✓ **High in proteins such as secretory IgA and lactoferrin**
- ✓ Has higher concentrations of (white blood cell provides immune protection), sodium, potassium, and chloride than mature milk.

TABLE 6.2 ■ Compositions of 100 mL colostrum (days 1–5 postpartum) and mature milk (day 15 postpartum)

CONTENTS	COLOSTRUM	MATURE MILK
Calories (kcal)	55	67
Fat (g)	2.9	4.2
Lactose (g)	5.3	7.0
Total protein (g)	2.0	1.1
Secretory IgA	0.5 ^a	0.1
Lactoferrin	0.5	0.2
Casein	0.5	0.4
Calcium (mg)	28	30
Sodium (mg)	48	15
Vitamin A (μ g retinol equivalents)	151	75
Vitamin B ₁ (μ g)	2	14
Vitamin B ₂ (μ g)	30	40
Vitamin C (μ g)	6	5

^a Concentration is considerably higher at 1–3 days postpartum than at days 4 and 5.

SOURCE: Adapted from Prentice A. Constituents of human milk. *Food and Nutrition Bulletin* 17(4), The United Nations University Press. December 1996.⁹

Human Milk Composition...

Water

- Breast milk is isotonic like plasma (has balanced concentration of salts and minerals).
 - Breast feeding babies do not need water or other fluids to maintain hydration, even in hot climates.

Energy

- ✓ ~0.65 kcal/mL
- ✓ Lower in calories than human milk substitute (HMS)

Human Milk Composition...

Protein (Casein, Whey Proteins)

- The protein content of mature human milk is relatively low compared to cow's milk.
- Casein
 - Main protein in mature human milk
 - Facilitates calcium absorption
- **Whey proteins**
 - Soluble protein that precipitates by acid or enzyme
 - Some mineral-, hormone-, and vitamin-binding proteins are part of whey
 - Includes lactoferrin (an iron carrier)

Human Milk Composition...

Lipids & Trans Fatty Acids

-  Lipids provide half of the energy of human milk.
 - ✓ Profile reflects dietary intake of mother
- DHA (docosahexaenoic acid)
 - ✓ Essential for retinal development
 - ✓ Associated with higher IQ score
- **Trans fatty acids**
 - Present in human milk from maternal diet

Cholesterol

- component of all cell membranes, and needed for growth and replication of cells.
- Higher in human milk than HMS
- Early consumption of cholesterol through breast milk appears to be related to lower blood cholesterol levels later in life

Human Milk Composition...

Carbohydrates

❖ Lactose

- ✓ dominant carbohydrate in human milk
- ✓ enhances calcium absorption.

❖ Oligosaccharides

- Medium-length carbohydrates
- Oligosaccharides and glycoprotein's are known collectively as the “bifidus factor”
- Prevent binding of pathogenic microorganisms, which prevents infection

Human Milk Composition...

Fat-Soluble Vitamins (Vitamin A, D, E & K)

- ✓ vitamin A-**Colostrum** has approximately **twice** the concentration of vitamin A as mature milk does.
- **Vitamin E**
 - Level linked to milk's fat content
- ✓ Level not adequate to meet needs of preterm infants
- **Vitamin K**
 - Infants who did not receive **K injection at birth** may be deficient

Water-Soluble Vitamins

- (Vitamin C, B6, riboflavin, niacin and biotin)
 - Content reflective of mother's diet or supplements
 - B6 (most likely to be deficient),
- **Vitamin B 12 & Folate**
 - Bound to whey proteins
 - Less influenced by maternal intake

Human Milk Composition...

❖ All minerals needed for newborn and infant growth are present in human milk

-Minerals (magnesium, calcium, iron, zinc) in human milk are bioavailable to the infant:

-Trace Minerals (copper, selenium, chromium, manganese, molybdenum, nickel & fluorine) are essential for growth and development.

bioavailability

	iron	zinc
Human milk	49%	41%
Cow milk	4%	31%
Fortified milk	10%	28%

Taste of Human Milk

- ❖ Flavor of foods in mother's diet influences taste of breast milk
 - Infants seem more interested in mother's milk if flavor is new
 - Exposure to a variety of flavors may contribute to infant's interest and acceptance of new flavors in solid foods

Benefits of Breastfeeding

Benefits of Breastfeeding for Mothers

- ✓ return uterus to non pregnant size,
- ✓ minimizes maternal postpartum blood loss & to expel placenta.
- ✓ Natural contraceptive
- ✓ Psychological benefits (facilitate bonding with their infants).
- ✓ Help the mother to lose excessive weight.
- ✓ Reduce the risk of breast and ovarian cancer.
- ✓ Economic benefits, breast milk is (almost) free, available at all times.

Benefits of Breastfeeding

Breastfeeding Benefits for Infants

❖ Nutritional Benefits:

- ✓ human milk contains balanced nutrients.
- ✓ Meets the infant's needs without overloading the immature kidneys with nitrogen.
- ✓ Whey protein in human milk forms a soft, easily digestible curd.
- ✓ Provides essential FA like: long-chain PUFA (DHA)- development of the central nervous system.

Benefits of Breastfeeding

- **Immunological Benefits:** protect against infections.
 - Breast milk contains
 - cells (T- and B-lymphocytes), IgA, Bifidus factor, carrier proteins (lactoferrin, vitamin B12-binding protein, and corticoid-binding protein), and
 - enzymes (lysozyme, lipoprotein lipase, leukocyte enzymes) confer immunological benefits.
- **Reductions in acute & Chronic Illness** (celiac disease, inflammatory bowel disease, allergy and asthmatic disease, ear infection).
- **Reduce the risk of childhood Overweight**
- **Cognitive Benefits:** cognitive development increases with the duration of breastfeeding.

Benefits of Breastfeeding

- **Socioeconomic Benefits:**
 - Cost saving
 - Decrease in medical care .
 - Environmental savings
 - Reduce the time lost due to infant morbidity and mortality.
 - Improve productivity.

The Breastfeeding Infant

- Optimal duration of breastfeeding
 - 2 year or longer
 - exclusively for 6 months & best to breastfeed for 24 months

The Breastfeeding Infant

Breastfeeding positioning



Rugby hold



Lying Down



cradle



Cross cradle

Breastfeeding Attachment

Good attachment includes

- Baby mouth is **wide open**
 - More of the darker skin (areola) above the baby's mouth than below
 - Baby lower **lip is turned outwards**
 - Baby's **chin is touching** mothers breast
-
- ❖ The baby suckles with **short quick** movements at first, but changes the rhythm to a more **continuous deep suckling** as the milk flows.

The importance of good attachment and positioning

- ✓ Ensures Sufficient Milk Supply
- ✓ Ensures Efficient Milk Transfer
- ✓ Happy Contented Baby
- ✓ Prevents Mum Becoming Sore
- ✓ Relaxed Mum



The Breastfeeding Infant

- Identifying hunger and satiety
 - Hunger is signaled by infant bringing hands to mouth, sucking on them, & moving head from side to side
 - Crying is late sign of hunger
 - Allow infant to nurse on one breast as long as they want to ensure they get hindmilk with its high fat content that provides satiety
- Note: High lactose content of foremilk may cause diarrhea

Breast feeding difficulties

- **Sore nipe**

- Some early discomfort expected
- Check latch on technique
- Wash daily with mild soap
- Nipple guards
- Pumping only when very sore

- **Engorgement (breasts feel hard, swollen and heavy)**

- Occurs when breasts overfilled with milk
 - Baby may have difficulty latching on due to swelling
- Prevent by nursing frequently
- When occurs, try pumping until breasts softer and baby can latch on

Breast feeding difficulties

- **Infection/mastitis**
 - Painful, enlarged, hard area of breast
 - Sometimes red area notable
 - Fever/flu-like symptoms in mother
 - Seek help early!
 - Need antibiotics + pain relief (acetaminophen)
 - Keep nursing if possible

Energy and Nutrient Needs for Lactation

- ❖ The DRIs for normal-weight lactating women assume that the energy spent for milk production is 500 Kcal per day in the first 6 months and 400 Kcal afterward.
- ❖ there should be a substantially greater dietary energy and nutrient intake during lactation.

WHO/FAO recommendations:

- ✓ If a mother delivers 850 ml of milk per day for six months, around **550 kcal/day** of additional energy is required to make that milk.
- The average energy requirements for **moderately active** woman is **2,200 kcal/day**, the total recommendation becomes **2,750 kcal/day**.

Maternal Energy Balance and Milk Composition

- Protein–calorie malnutrition results in **energy deficiency** and can reduce the **volume of milk produced** but does not compromise the composition of the milk.
- **Inadequate maternal diet not affect the *quality of milk produced***, (at the expense of maternal stores).
 - **Moderate weight reduction can be achieved without compromising infant weight gain**
 - Diets should be formed around healthy dietary patterns
 - Provide a healthy assortment of nutrients at specified calorie levels for each stage of breastfeeding

Protein

- On the basis of an 850 ml output, assuming it contains 1.2gram of protein per 100 ml, an extra 24 gram/day of protein with a chemical score of 70gram/day was recommended for lactation by FAO/WHO.

Vitamins and Minerals Supplement

- ❖ Well-nourished breastfeeding women do not need routine vitamin or mineral supplementation.
- ❖ Instead, supplementation should target specific nutritional needs of individual women. :
 - ✓ In the cases of proven dietary inadequacies, **vegan women** require vitamin B12 supplementation.
 - ✓ When **exposure to sunlight** is considered inadequate, maternal vitamin D supplementation may be advisable.
 - ✓ Women who avoid **dairy products** completely should use calcium (1200 mg) and vitamin D (10 µg) supplements

- The requirements for most are similar to requirements during pregnancy; except an increased requirement for vitamin A, calcium, vitamin C, and zinc.

Weight Loss During Breastfeeding

- Most women do not reach prepregnancy weight by one year after birth
- Modest energy reductions do not decrease milk production
- Women who consumed fewer than 1500 Kcal tended to experience a decrease in milk volume.
- On the other hand women who consumed over 1500 Kcal per day experienced no decrease.